BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, SWAT

INTERMEDIATE (11TH CLASS) EXAMINATION, 2023

PHYSICS

Past Paper 2023 • Paper Code: 112301

Retyped faithfully from the original board paper — same questions, no changes.

SECTION-A

Time Allowed: 20 Minutes | **Maximum Marks:** 18

Note: 1) Attempting all MCQs is compulsory. This paper along with the OMR sheet must be returned to the superintendent after due time. 2) Fill the circle A B C D, which one is correct with blue or black ball point in separate OMR Sheet like ● 3) If more than one circle in the OMR sheet is filled then no credit will be given to such answer.

Q1. $Kgms^{-1}$ is equivalent to

(A) Ns
(B) Ns^{-1}
(C) $N^{-1}s$
(D) Ns^{-2}

Q2. The angular speed in rad/hr for daily rotation of our earth is?

(A) π/6
(B) π/12
(C) 2π
(D) 4π

Q3. When Fx=3N and F=5N then Fy=?

(A) 6N
(B) 4N
(C) 2N
(D) 0N

Q4. At terminal velocity, the acceleration of falling object is

(A) $0ms^{-2}$
(B) $1ms^{-2}$
(C) $-9.8ms^{-2}$
(D) $+9.8ms^{-2}$

Q5. The slope of x-axis or any line parallel to x-axis is

(A) Maximum
(B) Uniform
(C) 0
(D) Infinity

Q6. Two forces of magnitude 20N and 50N act simultaneously on a body. Which one of the following forces cannot be the resultant

(A) 20N
(B) 30N
(C) 40N
(D) 70N

Q7. The restoring force of SHM is maximum when particle

(A) Displacement is maximum
(B) Crossing mean position
(C) Half way
(D) At rest

Q8. The net force that acts on a 10N falling object, when it encounters 4N of air resistance is

(A) [illegible in source scan]
(B) 4N
(C) 6N
(D) 10 N

Q9. Which of the following pair of angles will have the same horizontal range.

(A) 30° and 70°
(B) 15° and 75°
(C) 25° and 70°
(D) 30° and 65°

Q10. For the constructive interference between two set of waves, the phase difference should be

(A) 0
(B) 2π
(C) 4π
(D) All of these

Q11. The time rate of change of linear momentum is equal to

(A) Force
(B) Impulse
(C) Inertia
(D) energy

Q12. Which one of the following properties proves the transverse wave nature of light

(A) Interference
(B) Refraction
(C) Polarization
(D) Diffraction

Q13. Which one is constant for satellite in orbit?

(A) Velocity
(B) K.E
(C) P.E
(D) Angular Momentum

Q14. Triple point of water is

(A) 273.16°C
(B) 273.16°F
(C) 273.16K
(D) 372.16K

Q15. The atmosphere is held to the earth by

(A) Winds
(B) Gravity
(C) Clouds
(D) Rotation of earth

Q16. If the total work done around a closed path is equal to zero, the field is said to be

(A) Uniform
(B) Variable
(C) Conservative
(D) None of them

Q17. The percentage uncertainty in the measurement of (3.76±0.25)m is

(A) 4 %
(B) 6.6 %
(C) 25 %
(D) 376 %

Q18. The angle formed by a complete circumference at the centre of the circle is

(A) π
(B) 2π
(C) 3π
(D) 4π

SECTION "B"

Time Allowed: 2 hours and 40 minutes | **Maximum Marks:** 40

QI. Attempt any Ten Parts out of the following. Each Part carries equal marks.

(i) Show that the following relations are dimensionally correct. (a) $E=mc^2$ (b) PE= mgh

(ii) Define the following terms: (a) Unit vectors (b) Equal vectors (c) Null vector (d) Resultant vector.

(iii) An aeroplane while travelling horizontallly , dropped a bomb when it was exactly above the target, the bomb missed the target. Explain.

(iv) When you release an inflated but untied ballon, why dose it fly across the room.

(v) What happens to K.E of a bullet when it penetrates into a target?

(vi) Write the rotional analogue of the following. (a) F=ma (b) S=vt (c) p=mv (d) Vf = Vi + at

(vii) Define the following terms in thermodynamics: System, Surroundings, Boundary, State variable.

(viii) A soap bubble looks black when it bursts. Why

(ix) Why dose the speed of a sound wave in a gas changes with temperature?

(x) Define the terms Viscosity and viscous drag for fluids.

(xi) Define and calculate escape velocity of a body on earth.

(xii) Write the dimensions of (a) Work (b) Power (c) Velocity (d) Force

(xiii) Give two examples in which resonance plays an important role.

SECTION "C"

Maximum Marks: 27

Note: Attempt any Three questions of the following. Each question carries equal Marks.

QIII.

(a) Show that the motion of simple pendulum is conditionally SHM.

(b) A spider swings in the breeze from silk thread with a period of 0.6s. How long is the spider's strand of silk?

QIV.

(a) State and explain the first law of thermodynamics.

(b) Find the speed of sound in a steel railway track, if the density of steel is $7800 kgm^{-3}$ and Elastic modulus is $2.0 \times 10^{11} Nm^{-2}$.

QV.

(a) Define conservative and non-conservative forces. Show that gravitational field is a Conservative field.

(b) A garden hose of inner radius 1.25cm carries water at 2.60m/s. The nozzle at the end has radius 0.30cm. How fast does the water emerge out through the nozzle?

QVI. Write notes on any two of the following. All carry equal marks. (i) Stockes law (ii) Conditions for interference (iii) Angular Momentum